NexusOS — Part 01 Engineering Report

Executive Summary

Part 01 (Foundation) establishes the production-ready engineering foundation for NexusOS. The milestone focused on project structure, backend foundation, frontend foundation, database infrastructure, documentation, validation, repository organization, and engineering workflow. All planned acceptance criteria for this milestone were satisfied and the milestone was frozen after successful validation.

Milestone Scope

• Backend foundation: FastAPI, SQLAlchemy, PostgreSQL, Redis, Alembic.

• Frontend foundation: React, TypeScript, Vite.

• Security foundation and tenant isolation.

• Health endpoints and application structure.

• Documentation reorganization with preserved Git history.

• Validation pipeline and production build verification.

Validation Results

Validation Item	Result
Backend tests	10/10 Passed
Frontend tests	7/7 Passed
ESLint	Passed (warnings only)
TypeScript	Passed
Production Build	Passed
Git History	Clean
Working Tree	Clean
Repository Sync	Local and remote synchronized

Engineering Decisions

• Repository documentation reorganized into archive, engineering, product, and reports.

• Git history preserved using file moves rather than delete/recreate operations.

• Documentation references updated after repository restructuring.

• Production monitoring intentionally deferred until functional application modules exist.

• Milestone frozen only after achieving a clean working tree.

Known Observations

Frontend lint reported four react-refresh warnings. These are development-time recommendations and did not affect production builds or deployment readiness. Backend emitted deprecation warnings from dependencies running on Python 3.14; these do not block the milestone but should be monitored during future dependency upgrades.

Freeze Decision

Status: APPROVED
Milestone: Part 01 — Foundation
Decision: Frozen
Next Authorized Milestone: Part 02

Engineering Sign-off

This milestone satisfies its planned objectives. Validation, repository integrity, documentation, and deployment verification were completed successfully. The repository was synchronized with the remote and reached a clean working tree before freeze. Part 01 is officially complete.